

MBI Signal Logger Visual Guide

Application, system and information-flow maps

Portfolio context

Developed during my IT internship at Murphy Ben International (MBI) as an internship/internal experimental project for collecting and analyzing broadcast signal reports. I was responsible for the design and implementation of the application, working under the guidance of a senior member of the IT team.

This repository is a portfolio representation of the project. Sensitive configuration, internal information, production data, credentials, and company-specific deployment details have been excluded.

1. Application map

User	Interface	Main outcome
Listener or fan	Public Logger	Submit a quick location-aware reception report
Field personnel	Field Logger	Capture a detailed observation with GPS and technical context
Administrator	Admin Center	Review, configure, map, manage and export reports

2. System map

```
[ Listener ] [ Field staff ] [ Administrator ]
| | |
+----- Browser interfaces -----+
| HTTPS 443
| Microsoft IIS ]
| |
Static files /api/* via ARR
|
[ Node.js + Express ]
|
[ Field | Public | Config | Audit ]
```

3. Public page map

Area	What the user does
Introduction	Reads the purpose and begins the report
Location	Allows GPS and checks that a position is available
Station context	Selects station and optional channel
Reception choices	Selects quality, stability and service condition
Observation	Adds optional supporting detail
Summary and submit	Reviews and sends the report

4. Field page map

Area	Purpose
New Incident	Primary report-entry workspace
GPS controls	Capture position and accuracy
Station and channel	Identify the broadcast service being checked
Reception and technical fields	Record observations and optional readings
History	Review recent Field reports
Live Map	Map incident and current device locations
Offline tools	Queue or manage reports when connectivity is limited

5. Admin page map

Group	Areas
Overview	Dashboard and recent incidents
Applications	Field App and Public App workspaces
Management	Stations and channels
Operations	Incidents, history, maps and exports
Administration	Users, permissions, settings, GPS and audit
Configuration	Design, menus, components, forms, content and dashboard layout

6. Data and reporting map

```
Public submission -> public_incidents.json --+
+> combined Admin analysis -> CSV / Print / KML
Field submission -> incidents.json -----+
```

7. Visual verification checklist

- The correct app title and purpose are visible.
- Required fields are clearly marked.
- GPS state and accuracy are understandable.
- Selected reception choices are visually distinct.
- Submission success includes the generated incident ID.
- Admin filters match the records shown on maps and exports.